



Retrieving Data from TCGA via the GDC Portal

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Learning Objectives

- Understand the GDC portal structure and core navigation tools.
- Learn how to locate and filter TCGA datasets using the Repository and Cohort Builder.
- Use the GDC Data Transfer Tool to download data via UUID and manifest files.

GDC Portal: Recap

- **The Genomic Data Commons (GDC):** A unified platform for storing and accessing cancer genomics data.
- Primary source for **TCGA**, **TARGET**, and **GTEx** projects.
- Built by the NCI, provides both **open** and **controlled** access tiers. **Link:** <https://portal.gdc.cancer.gov>

Key Sections of the **GDC Portal**

- **Repository:** Search and filter for files by project, data type, workflow.
- **Cohort Builder:** Create and save cohorts based on clinical, demographic or molecular filters.

Key Sections of the **GDC Portal**

- **Analysis Center:** Interactive view of mutations, gene expression, sequencing reads by cancer type.
- **Submission:** For uploading your own research data (not focus today).

Analysis Center

- Explore data **visually** by gene, case or project.
- Mutation frequency, expression boxplots, CNV, survival plots.
- Link directly to files from any plot.

The Cohort Builder

- Define **patient-level filters**: cancer stage, age, vital status, etc.
- Save **filtered cohort** for file-level querying
- **Link to Repository** view to explore and export data

Using the **Repository** to Find TCGA Data

1. Go to <https://portal.gdc.cancer.gov/repository>
2. Filter by:
 - **Project** (e.g., TCGA-BRCA)
 - **Data category** (e.g., Transcriptome Profiling)
 - **Data type** (e.g., Gene Expression Quantification)
 - **Workflow** (e.g., HTSeq - FPKM)
3. View matching files and sample IDs

Download Options on GDC

- **Direct download** via the portal (small files, few samples).
- **Manifest file** download for large datasets.
- Use of **GDC Data Transfer Tool** for fast, resumable downloads.

What is a **Manifest File**?

- A **structured text file** listing all file **UUIDs** and **metadata**.
- Generated by clicking **“Add to Cart”** → **“Download Manifest”**.
- Used as input to GDC Data Transfer Tool.

Using the GDC Data Transfer Tool

- **Install** from: <https://gdc.cancer.gov/access-data/gdc-data-transfer-tool>.
- Open **terminal** (Linux/Mac) or **command prompt** (Windows) (Ensure zip file is in current working directory).
- Run: `unzip gdc-client_2.3_Ubuntu_x64-py3.8-ubuntu-20.04.zip`

Using the GDC Data Transfer Tool

- `chmod +x gdc-client`
- `sudo mv ~/path/to/gdc-client
/usr/local/bin/`
- `gdc-client download <file1 UUID>
<file2 UUID>`

Using the GDC Data Transfer Tool

- **Downloading data using UUIDs:**

```
gdc-client download file1 UUID file2
```

```
UUID
```

- **Downloading data using a manifest file:**

```
gdc-client download -m
```

```
/Users/PC/Downloads/gdc_manifest_file_name.txt
```

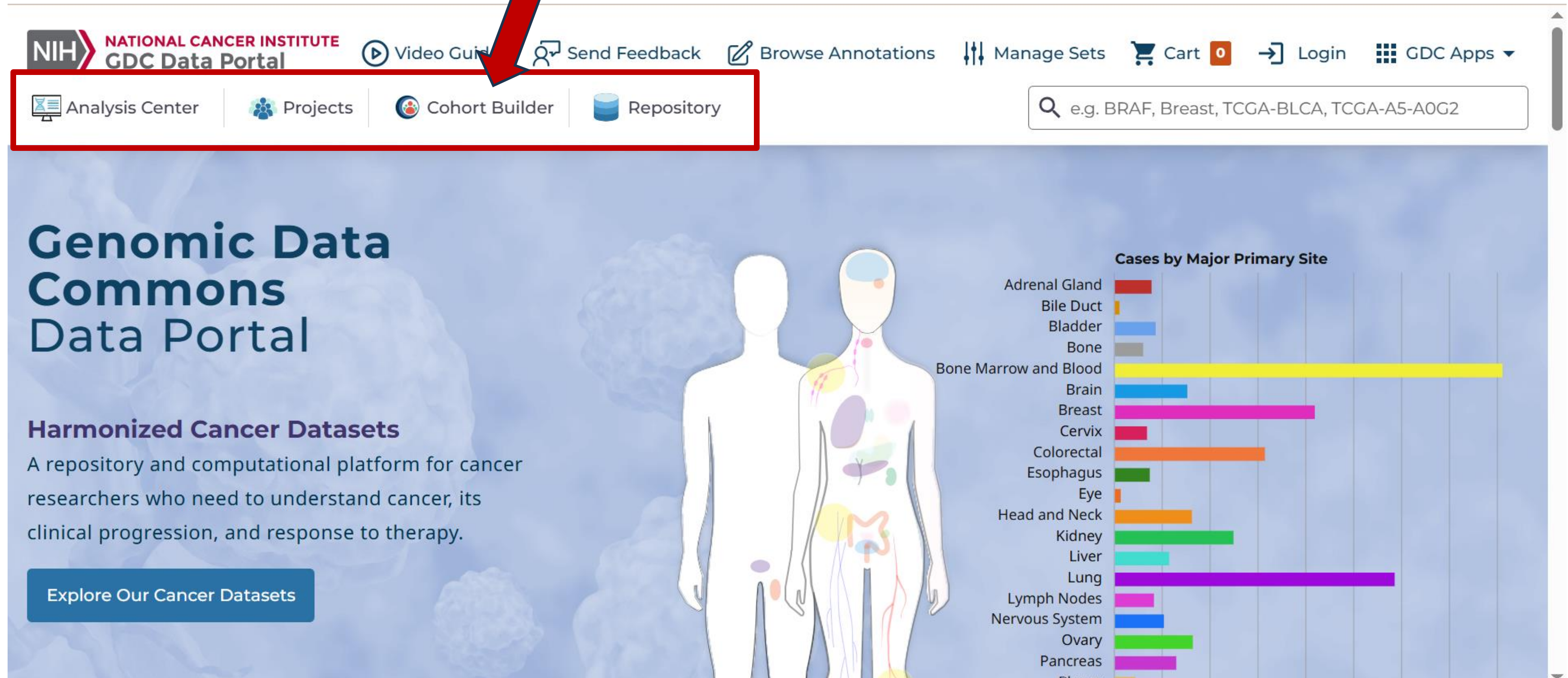

Best Practices for GDC Downloads

- Always double-check the workflow and **reference genome**.
- Use **manifest + gdc-client** for anything >10 files.
- Organize by sample/project folder.
- Track your token usage and expiry if accessing controlled files.

TASK...

Task: Retrieve CNV data from
the TCGA-LIHC project using
GDC portal

Key Sections of the GDC Portal



NIH NATIONAL CANCER INSTITUTE GDC Data Portal

Video Guide Send Feedback Browse Annotations Manage Sets Cart 0 Login GDC Apps

Analysis Center Projects Cohort Builder Repository

Genomic Data Commons Data Portal

Harmonized Cancer Datasets
A repository and computational platform for cancer researchers who need to understand cancer, its clinical progression, and response to therapy.

[Explore Our Cancer Datasets](#)

Cases by Major Primary Site

Primary Site	Relative Case Count (approximate)
Adrenal Gland	Low
Bile Duct	Low
Bladder	Low
Bone	Low
Bone Marrow and Blood	High
Brain	Low
Breast	High
Cervix	Low
Colorectal	High
Esophagus	Low
Eye	Low
Head and Neck	Low
Kidney	Low
Liver	Low
Lung	High
Lymph Nodes	Low
Nervous System	Low
Ovary	Low
Pancreas	Low
Prostate	Low

Key Sections of the GDC Portal

'Cohort Builder'

NIH NATIONAL CANCER INSTITUTE GDC Data Portal Video Guides Send Feedback Browse Annotations Manage Sets Cart 0 Login GDC Apps

Analysis Center Projects Cohort Builder Repository

Search: e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2

Save Cohort

Unsaved_Cohort Cohort not saved

Save Save As

PRIMARY SITE liver and intrahepatic cholangiocarcinoma PRIMARY DIAGNOSIS adenocarcinoma, nos PROGRAM TCGA

COHORT BUILDER Search

General

Demographic

General Diagnosis

Program	Cases
☐ MATCH	1 (0.00%)
☒ TCGA	14 (0.03%)

Project	Cases
☐ TCGA-CHOL	2 (0.00%)
☐ TCGA-LIHC	12 (0.03%)

Disease Type	Cases
☐ adenomas and adenocarcinomas	14 (0.03%)

Key Sections of the GDC Portal

'Cohort Builder'

NIH NATIONAL CANCER INSTITUTE
GDC Data Portal

Video Guides Send Feedback Browse Annotations Manage Sets Cart 0 Login GDC Apps

Analysis Center Projects Cohort Builder Repository

Search e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2

COHORT BUILDER

Search

- General
- Demographic**
- General Diagnosis
- Disease Status and History
- Disease Specific Classifications
- Treatment
- Exposure

Gender	Cases
☐ female	2 (0.00%)
☐ male	7 (0.02%)

Race	Cases
☐ asian	2 (0.00%)
☐ white	7 (0.02%)

Ethnicity	Cases
☐ not hispanic or latino	8 (0.02%)
☐ not reported	1 (0.00%)

Age at Diagnosis	Cases
From ≥ > Min: 0 yea	
To ≤ < Max: 90 y	
Apply	

Vital Status	Cases
☐ alive	7 (0.02%)
☐ dead	2 (0.00%)

Key Sections of the GDC Portal

'Cohort Builder'

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 **NATIONAL CANCER INSTITUTE**
GDC Data Portal

 Video Guides

 Send Feedback

 Browse Annotations

 Manage Sets

 Cart **0**

 Login

 GDC Apps ▾

 Analysis Center

 Projects

 Cohort Builder

 Repository

Treatment

Exposure

Biospecimen

Molecular Filters

Available Data

Custom Filters

Workflow Type	
Name ▲	Cases ▴▾
☐ ABSOLUTE LiftOver	13 (0.03%)
☐ Aliquot Ensemble So...	14 (0.03%)
☐ Arriba	12 (0.03%)
☐ ASCAT2	13 (0.03%)
☐ ASCAT3	13 (0.03%)
☐ AscatNGS	1 (0.00%)
+ 26 more	

Data Format	
Name ▲	Cases ▴▾
☐ bam	14 (0.03%)
☐ bcr biotab	14 (0.03%)
☐ bcr omf xml	14 (0.03%)
☐ bcr ssf xml	14 (0.03%)
☐ bcr xml	14 (0.03%)
☐ bedpe	14 (0.03%)
+ 9 more	

Platform	
Name ▲	Cases ▴▾
☐ affymetrix snp 6.0	13 (0.03%)
☐ illumina	14 (0.03%)
☐ illumina human met...	13 (0.03%)
☐ rppa	8 (0.02%)

Access	
Name ▲	Cases ▴▾
☐ controlled	14 (0.03%)
☐ open	14 (0.03%)

Key Sections of the GDC Portal

'Cohort Builder'

General

Demographic

General Diagnosis

Disease Status and History

Disease Specific Classifications

Treatment

Exposure

Biospecimen

Data Category	
Name ▲	Cases ▼
☒ copy number vari...	9 (0.02%)
show less	

Workflow Type	
Name ▲	Cases ▼
☐ AscatNGS	1 (0.00%)
☐ GATK4 CNV	9 (0.02%)

Access	
Name ▲	Cases ▼
☒ open	9 (0.02%)

Data Type	
Name ▲	Cases ▼
☐ Allele-specific Copy ...	1 (0.00%)
☐ Copy Number Segm...	9 (0.02%)
☐ Gene Level Copy Nu...	1 (0.00%)

Data Format	
Name ▲	Cases ▼
☒ tsv	1 (0.00%)
☒ txt	9 (0.02%)

Experimental Strategy	
Name ▲	Cases ▼
☒ Genotyping Array	0 (0.00%)
☒ WGS	9 (0.02%)

Platform	
Name ▲	Cases ▼
☐ affymetrix snp 6.0	13 (0.03%)
☒ illumina	9 (0.02%)

Key Sections of the GDC Portal

'Cohort Builder'

NIH NATIONAL CANCER INSTITUTE
GDC Data Portal

Video Guides Send Feedback Browse Annotations Manage Sets Cart 0 Login GDC Apps

Analysis Center Projects Cohort Builder Repository

e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2

COHORT BUILDER

Search

- General
- Demographic
- General Diagnosis
- Disease Status and History
- Disease Specific Classifications
- Treatment**
- Exposure

Best Overall Response	
Name ^	Cases ^
No data for this field	

Therapeutic Agents	
Name ^	Cases ^
☐ fluorouracil	1 (0.00%)
☐ oxaliplatin	1 (0.00%)

Treatment Intent Type	
Name ^	Cases ^
No data for this field	

Treatment Outcome	
Name ^	Cases ^
No data for this field	

Treatment Type	
Name ^	Cases ^
☐ pharmaceutical ther...	9 (0.02%)
☐ radiation therapy, nos	9 (0.02%)
☐ surgery, nos	8 (0.02%)

Key Sections of the GDC Portal

'Cohort Builder'

NIH NATIONAL CANCER INSTITUTE GDC Data Portal

Video Guides | Search | Feedback | Browse Annotations | Manage Sets | Cart 0 | Login | GDC Apps

Analysis Center | Projects | Cohort Builder | Repository

Search: e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2

Unsaved_Cohort

Cohort not saved

Save

Save As

PRIMARY SITE: liver and intrahepati... x

PRIMARY DIAGNOSIS: adenocarcinoma, nos x

PROGRAM: TCGA x

COHORT BUILDER

Search

General

Demographic

General Diagnosis

Program	Cases
☐ MATCH	1 (0.00%)
☒ TCGA	14 (0.03%)

Project	Cases
☐ TCGA-CHOL	2 (0.00%)
☐ TCGA-LIHC	12 (0.03%)

Disease Type	Cases
☐ adenomas and aden...	14 (0.03%)

Key Sections of the GDC Portal

The screenshot shows the GDC Data Portal Cohort Builder interface. A modal titled "SAVE COHORT" is open, prompting the user to "Provide a name to save the cohort." The "Name" field contains "TCGA-Adenocarcinoma" and has a character limit of "Maximum 100 characters". A red arrow points to the text input field. The background shows the "Unsamed_Cohort" section with filters for "PRIMARY SITE" (liver and intrahepati...) and "PRIMARY DIAGNOSIS" (adenocarcinoma, nos...). The "COHORT BUILDER" section is visible at the bottom, showing filters for "Program", "Project", and "Disease Type".

SAVE COHORT

Provide a name to save the cohort.

Name *

TCGA-Adenocarcinoma

Maximum 100 characters

Cancel Save

COHORT BUILDER

General

Demographic

General Diagnosis

Program	Cases
☐ MATCH	1 (0.00%)
☒ TCGA	14 (0.03%)

Project	Cases
☐ TCGA-CHOL	2 (0.00%)
☐ TCGA-LIHC	12 (0.03%)

Disease Type	Cases
☐ adenomas and aden...	14 (0.03%)

Key Sections of the GDC Portal

The screenshot displays the GDC Data Portal interface. At the top, the NIH National Cancer Institute GDC Data Portal logo is on the left, and navigation links for Video Guides, Send Feedback, Browse Annotations, Manage Sets, Cart (0), Login, and GDC Apps are on the right. Below this, a search bar contains the text "e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2". The main navigation bar includes links for Analysis Center, Projects, Cohort Builder, and Repository. A red arrow points to the Cohort Builder section, which is highlighted with a red border. The Cohort Builder section shows a list of filters for the TCGA-Adenocarcinoma cohort, including Primary Site (liver and intrahepatic), Primary Diagnosis (adenocarcinoma, nos), Program (TCGA), Data Category (copy number variation), Experimental Strategy (WGS, Genotyping Array), Data Format (tsv, txt), Platform (illumina), and Access (open). A "Changes not saved" message is visible above the filters. The bottom of the screen shows a "COHORT BUILDER" tab and a search bar.

NIH NATIONAL CANCER INSTITUTE
GDC Data Portal

Video Guides Send Feedback Browse Annotations Manage Sets Cart 0 Login GDC Apps

Analysis Center Projects Cohort Builder Repository

Search: e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2

TCGA-Adenocarcinoma

Changes not saved

TCGA-Adenocarcinoma Clear All

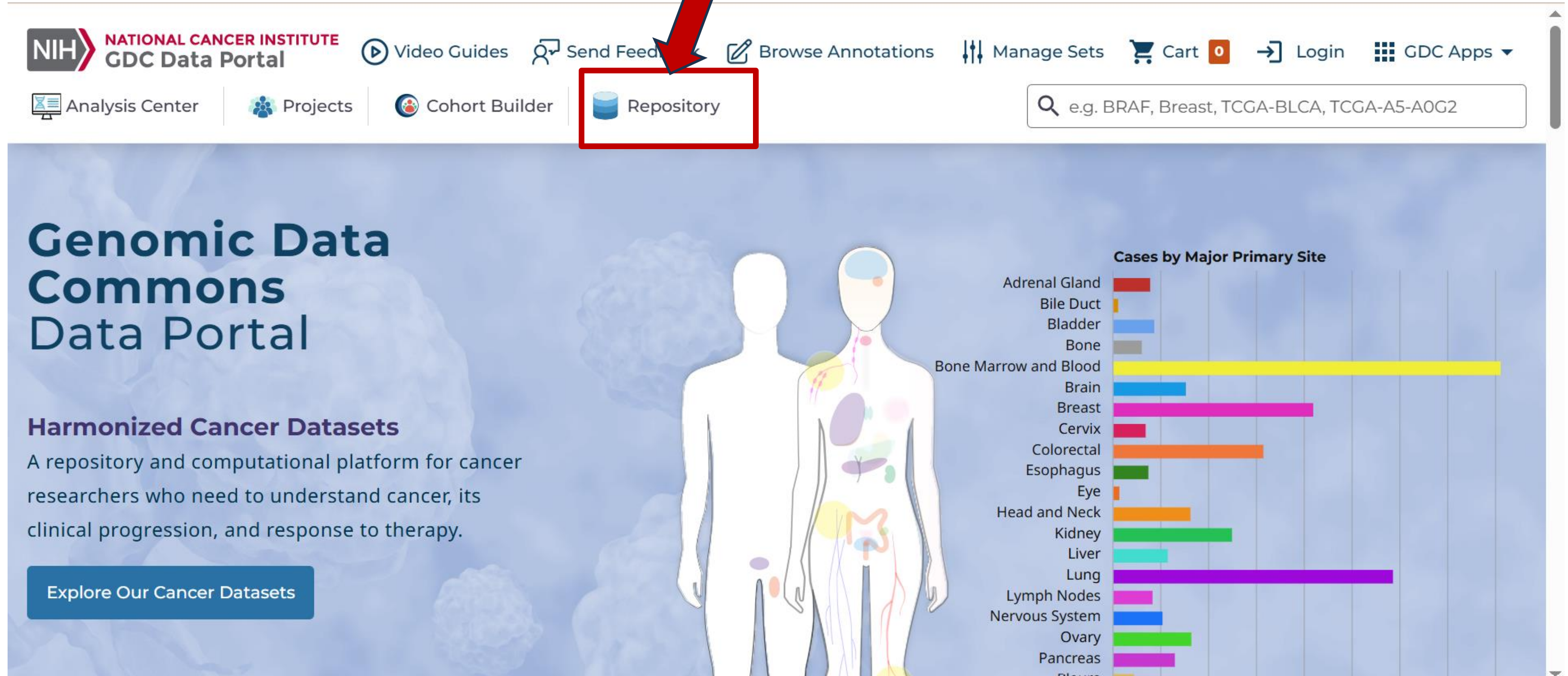
PRIMARY SITE liver and intrahepati... PRIMARY DIAGNOSIS adenocarcinoma, nos PROGRAM TCGA

DATA CATEGORY copy number variation EXPERIMENTAL STRATEGY WGS Genotyping Array DATA FORMAT tsv txt

PLATFORM illumina ACCESS open

COHORT BUILDER Search

Key Sections of the GDC Portal



The screenshot shows the GDC Data Portal homepage. A red arrow points to the 'Repository' link in the navigation bar, which is highlighted with a red box. The navigation bar includes links for Video Guides, Send Feedback, Browse Annotations, Manage Sets, Cart, Login, and GDC Apps. Below the navigation bar, there are links for Analysis Center, Projects, Cohort Builder, and Repository. A search bar is also present with the text 'e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2'.

Genomic Data Commons Data Portal

Harmonized Cancer Datasets
A repository and computational platform for cancer researchers who need to understand cancer, its clinical progression, and response to therapy.

[Explore Our Cancer Datasets](#)

Cases by Major Primary Site

Primary Site	Cases
Adrenal Gland	1
Bile Duct	1
Bladder	1
Bone	1
Bone Marrow and Blood	1
Brain	1
Breast	1
Cervix	1
Colorectal	1
Esophagus	1
Eye	1
Head and Neck	1
Kidney	1
Liver	1
Lung	1
Lymph Nodes	1
Nervous System	1
Ovary	1
Pancreas	1
Prostate	1

Key Sections of the GDC Portal

The screenshot shows the GDC Portal interface. A red arrow points to the left side pane, which contains filters for Data Format, Workflow Type, and Platform. Another red arrow points to the center pane, which displays a table of files in the cart. A third red arrow points to the top navigation bar. The interface includes a header with the NIH logo, navigation links, a search bar, and a summary of the total files, cases, and size. The left side pane has three sections: Data Format (tsv), Workflow Type (STAR - Counts), and Platform. The center pane shows a table with columns: Cart, Access, File Name, Cases, and Project. The table lists four files, all with 'Open' access and '1' case, from the 'HCTMI-CMD' project.

Left side pane

Center pane

Header: NIH NATIONAL CANCER INSTITUTE GDC Data Portal. Video Guides, Send Feedback, Browse Annotations, Manage Sets, Cart 0, Login, GDC Apps.

Navigation: Analysis Center, Projects, Cohort Builder, Repository.

Search: e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2

Filters: Collapse All, 4 Reset Filters, Sample Sheet, Metadata, Manifest, View Images, Add All Files to Cart.

Summary: TOTAL OF 25,504 FILES, 21,169 CASES, 107.85 GB

Buttons: + Add a Custom Filter, Remove All From Cart, JSON, TSV, Search.

Cart	Access	File Name	Cases	Project
	Open	3472a5d3-05ef-4b7b-a904-3476bd905aef.rna_seq.augmented_star_gene_counts.tsv	1	HCTMI-CMD
	Open	b437b05c-dd0d-4bf0-8a0c-d3f5b08fdf88.rna_seq.augmented_star_gene_counts.tsv	1	HCTMI-CMD
	Open	96d33b89-92f9-47dd-b70f-6c576d3309dd.rna_seq.augmented_star_gene_counts.tsv	1	HCTMI-CMD
	Open	2ab3664c-1dd8-4eff-bc41-cf262bcb6fcf.rna_seq.augmented_star_gene_counts.tsv	1	HCTMI-CMD

Key Sections of the GDC Portal

The screenshot displays the GDC Data Portal interface. At the top, the NIH National Cancer Institute GDC Data Portal logo is visible, along with navigation links for Video Guides, Send Feedback, Browse Annotations, Manage Sets, Cart (1), Login, and GDC Apps. Below the header, there are links for Analysis Center, Projects, Cohort Builder, and Repository. A search bar contains the text "e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-...".

The main content area shows a dataset titled "TCGA-Adenocarcinoma". A red box highlights a button labeled "8 CASES" with a downward arrow, indicating the number of cases in the dataset. A red arrow points to this button. Below the dataset title, there is a table of files. The table has columns for Name, Files, and a status column. The files listed include various genomic data files, such as "TCGA_LIHC.3171e434-f771-47a4-a7ed-d07b47e08657.wxs.Pindel.somatic_annotation.vcf.gz", "CHILE_p_TCGASNP_230_231_FFPE_N_GenomeWideSNP_6_D01_1147918.nocnv_grch38.seq.v2.txt", and "91512e3f-4731-4b3b-84c4-8fe05b4b40ba.wxs.aliquot_ensemble_masked.maf.gz".

On the left side, there is a sidebar with a "Data Category" section. It lists various data categories with their respective counts and percentages:

Name	Files
25x-150x	20 (0.00%)
not applicable	51 (0.00%)
Data Category	
biospecimen	44 (0.00%)
clinical	32 (0.00%)
copy number variation	110 (0.01%)
dna methylation	24 (0.00%)
proteome profiling	4 (0.00%)
sequencing reads	71 (0.01%)

Key Sections of the GDC Portal

Step 2: Click 'cart'

NIH NATIONAL CANCER INSTITUTE GDC Data Portal

Video Guides Send Feedback Browse Annotations Manage Sets **Cart 1** Login GDC Apps

Analysis Center Projects Cohort Builder Repository

Search: e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2

TCGA-Adenocarcinoma

8 CASES

Name	Files
☐ 25x-150x	20 (0.00%)
☐ not applicable	51 (0.00%)

Data Category

Name	Files
☐ biospecimen	44 (0.00%)
☐ clinical	32 (0.00%)
☐ copy number variation	110 (0.01%)
☐ dna methylation	24 (0.00%)
☐ proteome profiling	4 (0.00%)
☐ sequencing reads	71 (0.01%)

Step 1: Click the shopping cart

TCGA LIHC.3171e434-f771-47a4-a7ed-d07b47e08657.wxs.Pindel.somatic annotation.vcf.gz 1

CHILE_p_TCGASNP_230_231_FFPE_N_GenomeWideSNP_6_D01_1147918.nocnv_grch38.seq.v2.txt 1

91512e3f-4731-4b3b-84c4-8fe05b4b40ba.wxs.aliquot_ensemble_masked.maf.gz 1

CHILE_p_TCGASNP_230_231_FFPE_N_GenomeWideSNP_6_D07_1147862.grch38.seq.v2.txt 1

50cae4c7-fef8-40fa-a3ed-d14a34eb6103.rna_seq_augmented_star_gene_counts.tsv 1

CHILE_p_TCGASNP_230_231_FFPE_N_GenomeWideSNP_6_C12_1147980.nocnv_grch38.seq.v2.txt 1

Key Sections of the GDC Portal

 Download Cart ▾  Biospecimen ▾  Clinical ▾  Sample Sheet  Metadata  Remove From Cart ▾



HOW TO DOWNLOAD FILES IN MY CART?

Download Manifest:

Download a manifest for use with the [GDC Data Transfer Tool](#). The GDC Data Transfer Tool is recommended for transferring large volumes of data.

Download Cart:

Download Files in your Cart directly from the Web Browser.

Download Reference Files:

Download [GDC Reference Files](#) for use in your genomic data analysis.

FILE COUNTS BY AUTHORIZATION LEVEL

Level	Files	File Size
Authorized	1	27.22 kB
Unauthorized	0	0 B

FILE COUNTS BY PROJECT

Project	Cases	Files	File Size
TCGA-LIHC	1	1	27.22 kB

Key Sections of the GDC Portal

The screenshot displays the GDC Portal interface. At the top, the NIH National Cancer Institute GDC Data Portal logo is on the left, and navigation links for Video Guides, Send Feedback, Browse Annotations, Manage Sets, Cart (1), Login, and GDC Apps are on the right. Below the logo, links for Analysis Center, Projects, Cohort Builder, and Repository are visible. A search bar on the right contains the text 'e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2'. A red arrow points to the 'Download Cart' button in the 'TOTAL OF 1 FILE 1 CASE 27.22 KB' section. The dropdown menu for 'Download Cart' shows options for 'Manifest' and 'Cart'. Below this, a section titled 'LOAD FILES IN MY CART?' is visible. The 'FILE COUNTS BY AUTHORIZATION LEVEL' table shows 1 authorized file (27.22 kB) and 0 unauthorized files (0 B). The 'FILE COUNTS BY PROJECT' table shows 1 case and 1 file (27.22 kB) for TCGA-LIHC. The 'CART ITEMS' section at the bottom has buttons for JSON and TSV, and a search bar with the text 'TOTAL OF 1 FILE'.

NIH NATIONAL CANCER INSTITUTE
GDC Data Portal

Video Guides Send Feedback Browse Annotations Manage Sets Cart 1 Login GDC Apps

Analysis Center Projects Cohort Builder Repository

Search: e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2

TOTAL OF 1 FILE 1 CASE 27.22 KB

Download Cart Biospecimen Clinical Processing Metadata Remove From Cart

Manifest
Cart

LOAD FILES IN MY CART?

FILE COUNTS BY AUTHORIZATION LEVEL

Level	Files	File Size
Authorized	1	27.22 kB
Unauthorized	0	0 B

FILE COUNTS BY PROJECT

Project	Cases	Files	File Size
TCGA-LIHC	1	1	27.22 kB

CART ITEMS

JSON TSV

TOTAL OF 1 FILE Search

Key Sections of the GDC Portal

NIH NATIONAL CANCER INSTITUTE GDC Data Portal

[Video Guides](#) [Send Feedback](#) [Browse Annotations](#) [Manage Sets](#) [Cart 1](#) [Login](#) [GDC Apps](#)

[Analysis Center](#) [Projects](#) [Cohort Builder](#) [Repository](#)

Unauthorized 0 0 B

Search: e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2

CART ITEMS

[JSON](#) [TSV](#) TOTAL OF 1 FILE Search

Remove	Access	File Name	Cases	Project	Data Category	Data Format
	Open	CHILE_p_TCGASNP_230_231_FFPE_N_GenomeWideSNP_6_D07_1147862.grch38.seg.v2.txt	1	TCGA-LIHC	Copy Number Variation	TXT

Show 20 Entries Showing 1 - 1 of 1 file

Key Sections of the GDC Portal

The screenshot displays the GDC Portal interface. At the top, the NIH logo and 'NATIONAL CANCER INSTITUTE' text are visible. A blue header bar contains a file icon, the text 'FILE • CHILE_p_TCGASNP_230_231_FFPE_N_GenomeWideSNP_6_D07_1147862.grch38.seg.v2.txt', and a close button. Below this, a red 'Remove From Cart' button and a white 'Download' button are present. The main content area is divided into two sections: 'FILE PROPERTIES' and 'DATA INFORMATION'. The 'FILE PROPERTIES' section contains a table with details about the file, including its name, access level, UUID, data format, size, MD5 checksum, and project. The 'DATA INFORMATION' section contains a table with details about the data, including its category, type, experimental strategy, and platform. A red rectangle highlights the 'FILE PROPERTIES' and 'DATA INFORMATION' sections.

FILE PROPERTIES

Name	CHILE_p_TCGASNP_230_231_FFPE_N_Genome WideSNP_6_D07_1147862.grch38.seg.v2.txt
Access	open
UUID	1ec265e4-bbb8-4674-b00a-6477c9feef2d
Data Format	TXT
Size	27.22 kB
MD5 Checksum	c7ca09e6d8a445c4a20493c94dc54fe1
Project	TCGA-LIHC

DATA INFORMATION

Data Category	Copy Number Variation
Data Type	Copy Number Segment
Experimental Strategy	Genotyping Array
Platform	Affymetrix SNP 6.0

Conclusion

The GDC portal is your gateway to structured, high-quality cancer datasets. Mastering its features and knowing when to use **browser**, **manifest** or **API** sets you up for efficient, reproducible research.

Practice Task

Dataset: TCGA-COAD (Colon Adenocarcinoma) CNV &
RNA-Seq

Instructions: Go to:

https://portal.gdc.cancer.gov/analysis_page?app=CohortBuilder&tab=general

Practice Task - STEPS

1. Under Project, select: TCGA-COAD

2. Select the following under 'Available Data':

- a. Data Category: Copy Number Variation
- b. Data Type: Copy Number Segment
- c. Experimental Strategy: WGS

➤ **Question:**

- How many cases?

Practice Task - STEPS

3. Go to 'Repository'

4. Repeat steps 2a, b and c.

- **Additional* → Under Access (left side pane): Select 'Open'

➤ **Question:**

- What's the total number of files you arrived at?

Practice Task - STEPS

5. Select and **add** the following files to cart (at least 2).

- *TCGA-AA-3858-01A-01D-A91V-36.WholeGenome.RP-1657.cr.igv.reheader.seg.txt*
- *TCGA-A6-2682-01A-01D-A91U-36.WholeGenome.RP-1657.cr.igv.reheader.seg.txt*

➤ **Question:**

- What's the file format of both files?

Practice Task - STEPS

6. Download the manifest file.

7. Use the GDC transfer tool to download data from the manifest file

OR

8. Download zipped file by clicking 'Cart' (instead of 'manifest').

9. Unzip in your Windows/ OS file manager.

10. Take screenshot of your unzipped file folder.

Further Reading/ References

- **GDC Data Transfer Tool Docs:**

<https://gdc.cancer.gov/access-data/gdc-data-transfer-tool>

- **GDC API & Scripts:**

https://docs.gdc.cancer.gov/API/Users_Guide/Getting_Started/

- **YouTube:** “Downloading TCGA Data Using GDC Portal” by

NCI

Coming up next...

Utilizing and merging heterogeneous data:

- ✓ Clinical
- ✓ CNV
- ✓ RNA-Seq
- ✓ proteomics